

Wicked FBD: Applying Gravity



Version 4 These rules are final

Version 4: Clarified that flying monkeys need to stay outside the cage.
Each ruby slipper is counted.

Gameplay & Scoring, General:

Each full round of play is two minutes long, and comprises an initial 30s autonomous period that automatically rolls into the 90s “regular” RC period. Players may not touch their robots during game play.

No part of the robot can be moving prior to game play.

There are several scoring opportunities in this game:

- 1. Fields of Poppies**
- 2. Munchkinland Artifacts**
- 3. Train to Oz**
- 4. Hot Air Balloon**
- 5. Oz Head Levers**
- 6. Machinery’s Grimmerie (Gramoire, if we wanted to defend language, ugh.)**
- 7. Flying Monkeys**
- 8. Oz Multiplier**
- 9. Broom**

Players must remain on their respective sides of the game table, except in Munchkinland, in front of Oz’s head, or above Oz, at the broken window. These areas are common areas accessible to both players.

Players may start in one of three zones: On the train platform, towards the front of the poppy fields, or in Oz.

Course staff are empowered to make all scoring judgments and interpretation of rules. Remember, display unqualified kindness and respect in the face of fierce competition.

Fields of Poppies

Description: Collect poppies and bring them to Munchkinland.

Poppies	1 point per poppy in Munchkinland
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Notes:

1. Munchkinland is a common area inside the Yellow Brick Road and can be occupied by both players simultaneously.
2. Each poppy field contains 75 poppies, arranged randomly in the field.
3. Poppies in Munchkinland will be counted and scored at the end of game play. Points are awarded based on the number of respective player's color poppies in Munchkinland. Poppy points are not doubled for autonomous play. Any poppies within Munchkinland may be moved by either player.

Munchkinland Artifacts

Description: Points are awarded by placing the various artifacts (Tin Man's heart, Lion's Medal of Courage, Scarecrow's diploma, Dorothy's Ruby Slippers) onto the shelves. Extra points are awarded by matching the artifact with the correct character.

Artifact	Lower Shelf	Upper Shelf	Bonus (with character)
Heart	15	35	+5
Medal	15	35	+5
Diploma	15	35	+5
(Each) Ruby Slipper	15	35	+5



Notes:

1. Once placed, artifacts may not be removed by the other player. But artifacts may be picked up and moved by the original player in an effort to acquire a higher score. Note that the score will be based on the artifact's location at the end of game play.
2. To be counted, no part of a robot may be touching an artifact at the end of game play.
3. Artifacts are counted based on their location at the end of game play and are counted once and only once. They will be scored once, based on how they arrived at their positions – either during autonomous play or during RC play.

Train to Oz

Description: Points are awarded based on the distance traveled on the train to Oz.

Zone
First zone: 5 points
Second zone: 9 points
Third zone: 11 points
Fourth zone: 13 points



Notes:

1. Points are cumulative.
2. Points are awarded once and only once. Traveling back and forth repeatedly will not score multiple times.
3. If a distance is achieved during autonomous play, it is scored based on the cumulative zones reached during autonomous play. If a distance is obtained during regular play, the score is based on the cumulative zones reached during regular play. A single zone will not be scored for both play periods. For example, a train that travels to the first zone during autonomous play will be scored 10 points. Remaining at this elevation during the RC period does not count for additional RC points. However, should a player continue to travel during the RC period to the second mark, an additional 9 points will be awarded, for a total of 19 points.

Hot Air Balloon

Description: Raise the Hot Air Balloon to score points. Points are cumulative and awarded based on the maximum height achieved, multiplied by a “weight multiplier” based on the weight added to the balloon gondola. You’re being scored based on work! By the way, raising the balloon to its maximum height, hence escaping Oz, will make him mad!

Zone
First zone: 11 points
Second zone: 15 points
Third zone: 19 points
Fourth zone: 20 points

Weight Multiplier	
<2 lbs	1x
2-5 lbs	1.2x
5+ -8 lbs	1.3x
8+ lbs	1.4x



Notes:

1. Points are awarded once and only once. Traveling up and down repeatedly will not score multiple times.
2. If a height is achieved during autonomous play, it is scored based on the cumulative zones reached during autonomous play. If certain height is obtained during regular play, the score is based on the cumulative zones reached during regular play. A single zone will not be scored for both play periods. For example, an elevator that reaches the first zone during autonomous play will be scored 22 points. Remaining at this elevation during the RC period does not count for additional RC points. However, should a player continue to lift the balloon during the RC period to the second mark, an additional 19 points will be awarded, for a total of 31 points. In this scenario, if a player adds 3 lbs of artifacts, poppies, parts to the gondola, these 31 points will be multiplied by 1.2 = 37.2 points.

Oz Head Levers

Description: Move the levers to reveal Oz and score points. Each lever box contains three levers. Points are awarded based on alternatively moving a lever to its limits, i.e. back and forth.

Inner Lever	1.0 point per flip
Middle Lever	1.0 point per flip
Outer Lever	1.0 point per flip



Notes:

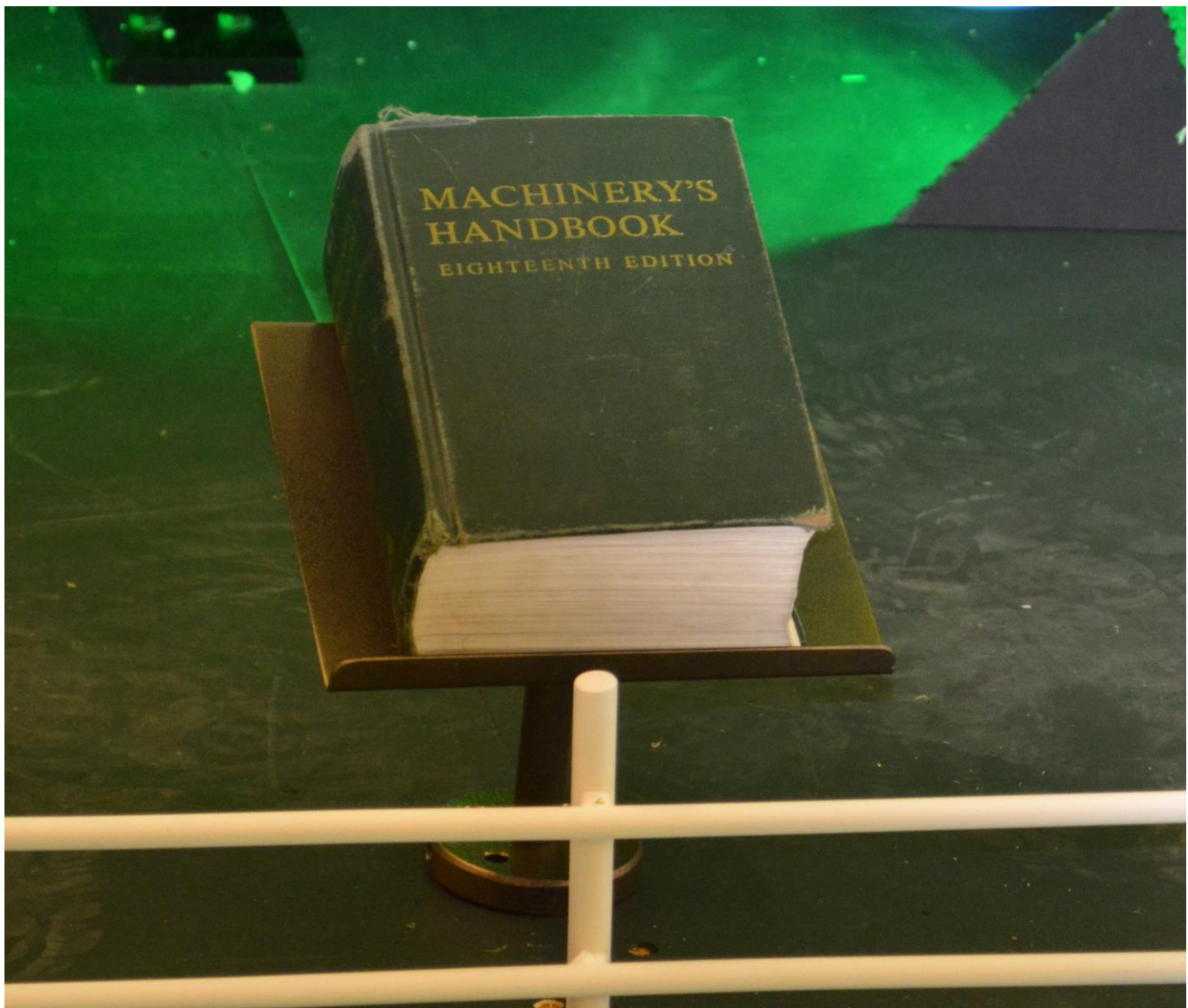
1. Consecutive points are awarded by alternating lever positions. For instance, a full push-pull-push-pull sequence for the inner lever would generate 4 points.

2. There is an (electronic) 1 second delay imposed between consecutive lever flips. Therefore, there is no advantage to press faster than 60 Hz.
3. Lever points are not doubled for autonomous play.

Machinery's Grimmerie

Description: The player who collects the Machinery's Grimmerie will earn the points. The Machinery's Grimmerie must be fully within a player's respective side of the game table to be scored.

Machinery's Grimmerie	23 points
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Flying Monkeys

Description: Retrieve the flying monkeys from the cage for points.



Scoring: Each flying monkey is worth 11 points. To score, a flying monkey must be fully outside the cage and remain outside for the remainder of game play. The cage is defined by the space between the yellow brick road and train ramps, behind the vertical surface defined by the front-most bars, in front of the blue Oz back wall, and below the plane of Oz. Set them free!

Oz Head Multiplier

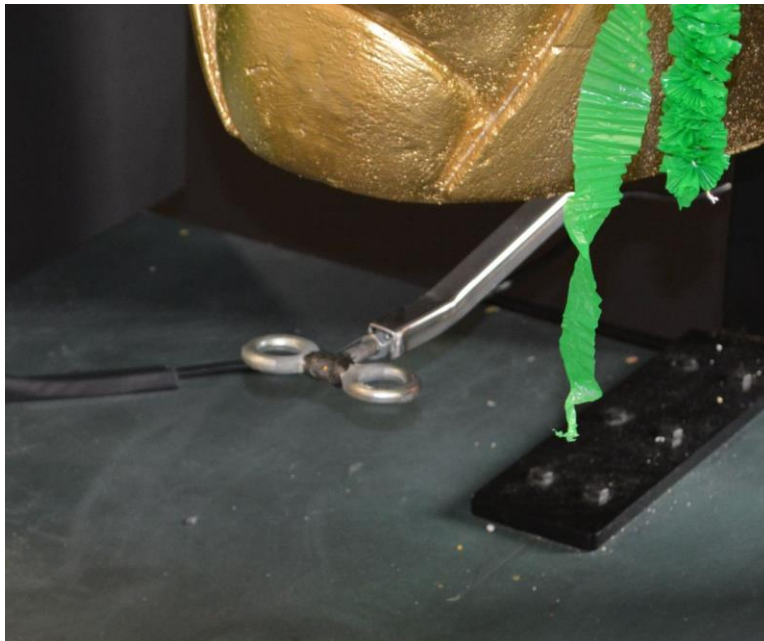
Scoring:

Oz's head may be tilted toward your side to obtain a multiplier for your points. The multiplier is determined by the maximum angle obtained. The highest multiplier value is applied to all points, and the multiplier may be obtained at any time during game play. The maximum multiplier obtained by each player will be applied once, and only once. Official referees, course instructors, and staff will determine which multiplier value is obtained.

Multipliers:

- Angle 1: 1.2x
- Angle 2: 1.5x
- Angle 3: 1.8x
- Angle 4: 2.0x

The angle zones will be indicated by a gauge or electronic display, and will be roughly equally distributed from zero tilt through a hard-stop position.



Notes:

1. The head is a pendulum and can swing to your opponent's side. Such swinging will not generate a multiplier for the opposing player. Judges will determine the player(s) responsible for generating the angle.
2. Players may not wedge any object or otherwise impede their opponent's ability to use the multiplier, except by virtue of actively attempting to move Oz themselves. Two players simultaneously, actively moving Oz is permitted. Placing a wedge, block, or "inactive" robot as a blocking strategy is in opposition to our principle of gracious professionalism, and not permitted. Don't be "that person" - ugh.

Broom

Description: Collect the broom for extra points. You know....Defying gravity and all that....

Broom	150 points
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Notes:

1. The area immediately above Oz's head is common area.
2. The Broom will be scored based on where it is at the end of game play. The broom must be completely on a player's half of the game table to be scored for that player.

Autonomous Period:

To encourage exploration and learning of electronics in 2.007, the Autonomous Period is 30 seconds. All points are doubled during the autonomous period, except poppy points. If neither competitor wishes to use the autonomous period, they can indicate to the referee and the contest will begin with the 90 second remote control phase.

Scoring Examples:

Game 1:

Blue pushes 12 poppies into Munchkinland, and then places the Tin Man's heart and a Ruby Slipper onto the second shelf. Blue then pushes ten more poppies into Munchkinland before driving up the Yellow Brick Road to pull the rope that lifts the hot air balloon to the second zone.

Blue's score: 22 [poppies] + 40 [Tin Man Heart, Top, correct character] 40 [Ruby Slipper, Top, correct charcter] + 26 [Balloon] = 128 points

Red took the train directly to Oz, and rode the balloon during RC play to the highest zone. Red weighed 6 lbs.

Red's score: $38 + 65 * 1.3 = 122.5$ points

What a round! Blue prevails!

Game 2:

Blue takes the Yellow Brick Road to Oz, flips all levers once, grabs Oz's head for a 2x multiplier, and gets the Machinery's Grimmerie. They attempted to lift the balloon by pulling the rope (look at the game table!), but only reached the second zone because their robot wheel fell off.

Blue's score: $(3 \text{ [all levers one flip]} + 23 \text{ [Machinery's Grimmerie]} + 26 \text{ [balloon]}) * 2.0 \text{ [Oz pull]} = 104$ points

Red pushed 5 poppies into Munchkinland and also drove up the Yellow-Brick Road. Their robot engaged two of the levers and flipped them repeatedly for nearly a minute then in the last seconds of the game, pulled the Oz multiplier to its maximum angle.

Red's score: $((5 \text{ [poppies]} + (2 * 25 \text{ times}) \text{ [levers]}) * 2.0 \text{ [Oz pull]}) = 110$ points

What an exciting round!! Red has the higher score and therefore wins.

Rules & Regulations

1. General Principles

- a. These rules are intended to create opportunities to learn engineering.
- b. Those things not specifically forbidden are allowed.

2. Timing

- a. Each round of the contest is 120 seconds long.
- b. For the first 30 seconds, no control signals may be sent to the robot. So that your robot can sense the start of the round, players will trigger the start of the autonomous round with their own light source – this may be a laser pointer, flashlight, phone, etc. Players may not touch the robot to initiate the autonomous period.
- c. If both players agree to not use the first 30 seconds, then the match will last 90 seconds.
- d. Robots may not be touched by players during game play.

3. Winning & Advancing

- a. There will be a single elimination tournament with 32 slots in the bracket. The winner of the tournament is the winner of the contest.
- b. In the tournament, the highest scoring robot in each match advances to the next round. In general, only one machine may advance. If there is a tie, it will be broken by weighing the two systems. The lighter machine prevails.
- c. There are two ways to earn a spot in the tournament bracket – 16 spots are given to the 16 highest players in a ladder, 16 more spots are awarded through a qualification tournament.
- d. The qualification tournament has only one round. Of those who win their match, the top 16 scorers advance into the single elimination tournament. The qualification tournament is run the night before the final tournament. Placement within this tournament bracket is determined by seeding scores. Seeding rounds will be held during the last week of lab time. In the seeding rounds, machines will run on the table unopposed.
- e. Ladder rounds can be held at any point on or after February 24, at any time the lab is open. A student establishes their position on the ladder by: 1) demonstrating to a section instructor, TA, or UA that they can score at least one point, and 2) placing their name on the list according to the points they receive. If a student ties with a score already on the ladder, the student is in the position below that score. The ladder is simply a list of students, ordered according to the points they can obtain during the semester. The ladder may include all students, but only the top 16 scorers secure their position for the final round of play, i.e. the second night of our festivities. The ladder will be closed end-of-day, Friday before Impound Week.

4. Control

- a. When possible, contestants must participate in controlling their own machines. Rarely, schedule conflicts may preclude a player from operating their machine, in which case, with their instructor's permission, another player can be designated.
- b. All control must be accomplished without contacting the robot.
- c. Control may be achieved via radios and, in addition, any another wireless device approved by their section instructor (e.g. TV remote control, laser pointer, recorder, accordion, bagpipes, cell phone, Xbee, Bluetooth, etc).
- d. A contestant may use their single lab issued six channel radio controller. However, contestants may add more degrees of control by using additional controllers such as a TV remote control, laser pointer, receiver, or playstation controller.
- e. Outside Controllers are permitted, but only to supplant the existing 6-channel controllers.
- f. A contestant may not deliberately interfere with the radio or other remote control of the opposing player.
- g. "Assistant drivers" may be used, though only one standard lab RC may be used.
- h. Contestants must wear safety glasses when in the vicinity of the table. Some prescription glasses are acceptable.

5. Robot Configuration

- a. Your entire robot must fit in a *Starting Volume* at the time of impounding and at the beginning of each match when set up on the table. The starting volume is 12" square and 16" tall. The box is "virtual" and has no ceiling or door. These dimensions are smaller than those of the lockers, so you should be able to fit the robot and also your tools and spare materials in your locker. This virtual starting box may be placed anywhere within the colored starting zone at the front of the tables. The lab-supplied box is used as a convenient gauge, and is not used to constrain your robot. That is, your robot must be within the stated virtual volume without anything holding it within those dimensions.
- b. Your entire robot must be made from lab supplied materials and components, fasteners and items in the stock shelves, approved batteries, and electronic components authorized by your section instructor. The tools in the tool kit are considered to be part of your robot kit and can therefore be part of your machine as long as safety and protection of the table are adequately considered in your design. Items that primarily serve to make robots look cool generally will be allowed including LEDs, seven-segment displays, and such items. There are no chemicals in the kit or supply cabinet aside from epoxy, cyanoacrylate adhesive, grease, and such things. For example, baking soda and vinegar would make a great balloon inflation means, but they are not in the kit and cannot be in your robot. Electric motors (or other actuators) taken from appliances (or other household or industrial equipment) are not allowed.
- c. Your "kit" includes up to 6 cubic inches of 3D printed parts. You are also limited to consuming up to 4 cubic inches of support material.
- d. You are limited to 1 hour total cut time throughout the semester on the water-jet. Use your time wisely.

2.007 Design & Manufacturing I – Spring 2025

- e. You may use any combination of four lab supplied gearhead motors AND any combination of four lab-supplied continuous rotation and position servos.
- f. Machine weight is limited to 12lbs. You should generally still attend to the weight of your machine as excess weight may lead to poor performance. Also, weight of your system will be used as a tie breaker.
- g. Energy may be stored in batteries, compressed air, elastic strain, and gravitational potential energy. Total stored energy may not exceed 50kJ. This limit will be enforced by the section instructors based on calculations in the lab notebooks. Compressed air may not exceed 50 psi gauge pressure and all air containers can add up to 2 liters maximum. Mechanisms using large amounts of rubber or springs must have adequate safety locks to reduce the chance of accidents. These locks may be removed once all contestants and onlookers are safely away from the device. Safety of all forms of energy storage will be at the discretion of the section instructors and judges.
- h. No boost converters are allowed. Voltage is limited to the 7.4V LiPO batteries. If 7.4 V is to be supplied to any electronics, it must be approved and checked by an instructor AND must be fused. The LiPo batteries are considered 800 mA-hr devices, regardless of their actual markings.
- i. The rechargeable AA -four packs that are typically used with the lab radio receivers are considered 500 mA-hr devices, regardless of the actual labeling on the batteries.
- j. Contestants will be responsible for charging their own batteries, compressed air containers, springs, rubber bands, etc.
- k. Your machine may be reconfigured between rounds. One reason to reconfigure is to accommodate the differences between the right and left sides of the field. You will know at least 5 minutes before you compete which side your machine will be assigned.
- l. You will have access to set up your machine(s) within the starting box prior to each round. You should be able to complete your set-up fully in fewer than three minutes.

6. Sporting Conduct & Safety - Gracious Professionalism

- a. Munchkinland and the areas in front of Oz's head and above Oz's head are considered common space and useable by both players.
- b. Crossing to the other player's side of the table is not permitted, except within common spaces.
- c. Damaging the contest table and or control equipment is strictly forbidden.
- d. In the case of destruction deemed by the judges to be accidental, the judges may permit repairs and a rematch.
- e. Contestants and spectators (i.e. any human beings) may not directly affect the motion of the machines or anything else on the table.
- f. It is permissible for a robot to reach outside the boundaries of the table during a round.
- g. Nets or entanglement devices are not permitted, but other defensive devices generally are permitted.
- h. NO DANGEROUS MACHINES. THE JUDGES' DECISIONS ON SAFETY MUST BE RESPECTED AND OBEYED PROMPTLY.

General questions may be asked of UA's, TA's, and Instructors; however, all officially sanctioned rule interpretations will be made by Council of Oz Elders.